

Case Study: Data-Driven Decision Making

Background

You are the newly appointed **Data Analyst** for “EduNation JEE Academy,” a nationwide network of JEE coaching centers operating in 30 cities across India. The institute caters to around 25,000 students annually across two main programs: **1-Year Course** (Class 12 students) and **2-Year Course** (Class 11 students). The CEO wants **centralized, data-driven insights** that can help the **Sales Head, Finance Head, and Academics Head** make informed decisions. You have been provided with **raw data from multiple sources** over the past two years.

Objective

You are expected to:

1. **Assimilate** the data from multiple files (Sales CRM exports, Finance records, Academic test results).
2. **Clean** the data to ensure accuracy, remove inconsistencies, and make it analysis-ready.
3. **Analyze** the data to find key trends, patterns, and correlations.
4. **Interpret** the findings in a way that supports decision-making for each department.
5. **Present** actionable **business insights** and recommendations to the CEO.

Available Data

You have been given three datasets:

1. Sales Data (Leads & Enrollments)

- Fields: Lead_ID, Date_of_Enquiry, City, Source (Social Media / Referral / Walk-in / School Seminar), Course_Type (1-Year / 2-Year), Conversion_Status (Converted / Not Converted), Date_of_Joining, Fees_Quoted
- Issues: Missing city names in some rows, inconsistent date formats, typos in course names, duplicated leads

2. Finance Data

- Fields: Student_ID, City, Course_Type, Total_Fees, Amount_Paid, Payment_Mode, Payment_Date, Outstanding_Amount
- Issues: Missing payment dates, negative outstanding values, mismatched student IDs with sales data

3. Academics Data

- Fields: Student_ID, City, Course_Type, Test_Date, Subject (Physics / Chemistry / Math), Marks_Obtained, Maximum_Marks

- Issues: Missing marks in some tests, inconsistent subject spellings, wrong maximum marks in a few entries

Tasks for the Candidate

A. Data Assimilation

- Merge all three datasets into a **single structured database** with a unique student identifier.
- Ensure all entries are aligned by City, Course_Type, and Student_ID.

B. Data Cleaning

- Handle missing values (justify chosen method).
- Correct inconsistent formats and spellings.
- Remove duplicates and resolve mismatched records between datasets.
- Verify data integrity (e.g., no negative outstanding amounts).

C. Analysis & Interpretation

1. **For Sales Head**
 - Identify top-performing lead sources and their conversion rates.
 - Compare enrollment trends across cities and course types.
 - Analyze seasonal patterns in enrollments.
2. **For Finance Head**
 - Identify fee collection efficiency by city and course type.
 - Highlight overdue payments and risk areas.
 - Analyze correlations between payment delays and dropouts.
3. **For Academics Head**
 - Track student performance trends across subjects and cities.
 - Identify weak subjects by city and batch.
 - Correlate academic performance with payment completion and attendance.

D. Business Insights for CEO

- Provide **3-5 strategic recommendations** that balance Sales growth, Finance stability, and Academic excellence.
- Suggest measurable KPIs for continuous monitoring.

Deliverables

- **Cleaned & Merged Dataset** (Excel/CSV)
- **Dashboard** (Excel/Power BI/Tableau) summarizing key insights
- **Presentation** (max 10 slides) for the CEO and department heads, with:
 - Key metrics & visualizations
 - Insights & implications
 - Recommended actions